

Elison Tuscano

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ABOUT ME

Senior Software Engineer with 5+ years of experience across full-stack development, distributed systems, cloud platform engineering, and SRE. Skilled in Python, React, and TypeScript, with experience designing scalable systems on AWS, automating infrastructure with Terraform and Kubernetes, and optimizing CI/CD pipelines. Engineered a production-grade generative AI solution on an HPC platform, including a Retrieval-Augmented Generation (RAG) pipeline on AWS Bedrock and LLM-powered troubleshooting agents. Proven track record of improving system reliability and reducing costs by \$270K+.

TECHNICAL SKILLS & CERTIFICATIONS

AI & GenAI: AWS Bedrock, Amazon Bedrock AgentCore, Retrieval-Augmented Generation (RAG), LLMs, LLM Agents, Model Context Protocol (MCP), Embeddings, Vector Search, Prompt Engineering

Languages & Frameworks: Python, Java, React, JavaScript, TypeScript, Node.js, Flask, Shell, SQL, NoSQL

Databases: PostgreSQL, Redis, MongoDB, DynamoDB, AWS RDS

Cloud Platforms: Amazon Web Services (AWS: EC2, Lambda, S3, Redshift, Glue, CodePipeline), Google Cloud (BigQuery, Data Studio), Microsoft Azure

DevOps: Docker, Kubernetes, Splunk, Grafana, Prometheus, Jenkins, Ansible, Linux, Terraform, Git

Big Data & Analytics: Tableau, BigQuery, TensorFlow, PyTorch, Apache Spark, Hadoop, Kafka, Airflow

Certifications: HashiCorp Terraform Associate (003)

PROFESSIONAL EXPERIENCE

Amazon – Software Development Engineer II **Nov 2024 – Present**

[React, TypeScript, Python, AWS, Redshift, QuickSight, Grafana, CloudWatch, Bedrock, MCP]

- Built a scalable, cloud-based High-Performance Computing (HPC) platform to execute heavy compute workloads like AI training, chip design and complex simulations, processing over 10 million jobs annually
- Developed LLM-powered troubleshooting and admin agents for HPC and DCV diagnostics and remediation, reducing SLA incidents by 20%, with agent audit logging and usage insights on QuickSight dashboards.
- Designed an MCP (Model Context Protocol) server framework exposing platform modules and REST APIs to LLM agents over local (SSH) and remote (Bedrock AgentCore) servers, secured with Cognito JWT and SigV4 IAM authentication.
- Implemented a production RAG application on AWS Bedrock that lets users query HPC job and simulation data in natural language, grounding answers for platform data to reduce support load.
- Architected a KPI analytics pipeline using AWS Glue, Redshift, and QuickSight, publishing platform metrics and saving \$100K+ in cost through data-driven insights.
- Engineered a filesystem monitoring service, providing dashboards and automated alerts that gave admins storage visibility and reduced costs by \$120K per year.
- Extended on-demand DCV based virtual desktop service by implementing idle-detection and automated termination workflows, notifying users of unused sessions and saving \$50K+ annually in compute costs.
- Engineered multi-region support for the HPC platform, enabling users to provision jobs in the optimal region, reducing compute costs by 30% through compute-aware allocation.
- Led and owned the end-to-end release management lifecycle, coordinating cross-functional teams to build, test, and roll out images ensuring zero-downtime upgrades across production clusters.

Visa Inc. – Senior Site Reliability Engineer (DevOps)

Jun 2022 – Aug 2024

[Docker, Red Hat, OpenShift, Kubernetes, Linux, Jenkins, Ansible, Splunk, Grafana, Prometheus]

- Automated vulnerability findings reporting, saving 8 hours of manual work weekly.
- Migrated client-facing environment (4000 TPS) from VMs to Docker containers with Kubernetes, improving infrastructure, deployment process and scalability.

- Created dynamic monitoring dashboards and alerts with Splunk, Prometheus, and Grafana, reducing SLA incidents by 30%.
- Designed a CI/CD pipeline on AWS using CodePipeline, CodeDeploy, S3, CodeArtifact, CloudFormation, and Terraform for over 10 applications.
- Led root-cause analysis on production incidents and built Python/Bash automation to remediate recurring failures, reducing manual toil and improving system stability.

Tata Consultancy Services – Software Engineer, Data Platform

Sep 2021 – May 2022

[Google Cloud (GCP), BigQuery, Data Studio, Python, Scala, Apache Spark, HDFS, Kafka, Airflow]

- Facilitated migration and upgrades of company's product data from Oracle to Google Cloud while ensuring 99.99% availability of applications.
- Designed and streamlined a data pipeline to extract data from multiple sources, perform transformations, manage data warehousing, and integrate with existing applications.
- Implemented Kafka for real-time data streaming and Spark MLlib for market analysis and data modeling.

Code Science Technologies Pvt Ltd – Full Stack Software Developer

Jun 2018 – Jun 2019

[AWS, Python, Flask, PHP, Laravel, React, Node.js, MySQL, DynamoDB, JavaScript]

- Architected scalable and cost-effective AWS solutions for companies resulting in 20% decrease in infrastructure cost while reducing downtime.
- Optimized database performance by implementing AWS RDS and DynamoDB, resulting in 25% improvement in query response time.

EDUCATION

Master's in Artificial Intelligence/Data Science Campbellsville University

Jul 2024 – Mar 2026

Master's in Computer Science University of Texas at Arlington, Texas

Aug 2019 – May 2021

Bachelor's in Computer Science University of Mumbai, Mumbai

Jun 2015 – May 2019